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JAIDEV EDUCATION SOCIETY'S

J D COLLEGE OF ENGINEERING AND MANAGEMENT

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(An Autonomous Institute, with NAAC "A" Grade)

Affiliated to DBATU, RTMNU & MSBTE Mumbai

Department of CSE

"A place to learn; A chance to grow"

2025-2026(Odd Semester)

MSE-1 EXAMINATION

Course: B tech in CSE **Year/Semester:** 3rd Semester (2nd Year)

Subject: Data Structure Using Python (DSUP) **Subject Code:** CS3T002

Max Marks: 20 **Date: 05/08/2025** **Duration: - 1 Hr.**

Instructions to the Students:

1. All the questions carry marks as indicate.
2. Assume suitable data whenever necessary.
3. Illustrate your answer whenever necessary with the help of neat sketches.

Students should be able to:

CO 1: Students will be able to understand the basic concept of data structures, time complexity and analyse the various sorting and searching algorithms by using python

CO 2: Students will be able to implement dynamic data structures like singly, doubly and circular linked list.

CO 3: Students will be able to Apply the different linear data structures like stack and queue to various computing problems.

CO 4: Students will be able to Implement different types of trees and apply them to problem solution.

CO 5: Students will be able to Demonstrate the representation of graphs and their applications in real life problem

SECTION A

1. **All Questions are compulsory.**
2. **Objective type question (Compulsory Question)**

Q. No.	Questions	Level/CO Mapped	Marks
Q.1	Choose which is the correct way to create a list in Python? A. list = (1, 2, 3) B. list = {1, 2, 3} C. list = [1, 2, 3] D. list = <1, 2, 3>	L3/CO1	1
Q.2	Show the output of the following code? list1 = [10, 20, 30, 40] print(list1[-2]) A. 10 B. 20 C. 30 D. 40	L3/CO1	1
Q.3	Select Which of the following is a mutable data structure in Python? A. Tuple B. List C. String D. All of the above	L1/CO1	1
Q.4	Select which condition must be satisfied to perform binary search? A. The list must be of odd length B. The list must be sorted C. The list must be numeric D. The list must not contain duplicates	L1/CO1	1

SECTION B

1. **All Questions are compulsory.**
2. **Solve answer in brief (two, three Sentence).**

Q.5	Define Data Structure and list it's Types?	L1/CO1	2
Q.6	Describe is a tuple in Programming, and how is it different from a List?	L1/CO1	2
Q.7	Discuss how to Create and Access Dictionary in Python?	L2/CO1	2
Q.8	Recall the concept of Stack and Queue?	L1/CO3	2

SECTION C

1. **Solve any two questions.**

Q.9	Apply the Linear Search Algorithm and Consider the array arr[] = {10, 50, 30, 70, 80, 20, 90, 40} and Search the key = 80 .	L3/CO1	4
Q.10	Difference between Binary and Linear Search .	L4/CO1	4
Q.11	Apply the Binary Search Algorithm and Consider the array arr[] = {10,12,,20,32,50,55,65,80,99} and Search the key = 12.	L3/CO1	4